ReliefLink: Master Capstone Defense & Video Demo Script

A Blockchain-Powered Parish Relief & Donation Management System • End-to-End Walkthrough

Target Presentation Duration: 8 – 12 Minutes

Target Audience: Capstone Defense Panel & Industry Evaluators

Demonstration Platforms: Flutter Mobile (iOS/Android) + React Admin Portal (Web)

Blockchain Core: Hyperledger Besu Private Ethereum Ledger (QBFT)

ACT I: Executive Introduction & The Core Problem

Presenter Objective: Capture panel attention within 60 seconds by articulating the systemic flaw in traditional charitable donations and introducing ReliefLink's dual web-mobile architecture backed by blockchain.

■■ LIVE SCREEN DEMO ACTION:

Display: Title slide on projector or ReliefLink Web Landing Page.
Action: Stand with posture, make eye contact with panelists, do not click yet.

■■ SPEAKER SPOKEN SCRIPT:

"Good day, distinguished members of the capstone panel. In church organizations and charitable foundations, trust is the primary currency. However, millions in humanitarian contributions face an accountability gap: once cash is placed in collection baskets or sent via e-wallet, donors rarely know when, where, or even if their money reaches actual victims.

*To solve this, we developed **ReliefLink**: an integrated ecosystem combining an automated donor mobile app, a prescriptive parish administrative dashboard, and a private **Hyperledger Besu Ethereum blockchain**. Today, we will demonstrate our live system across both mobile and web to prove that every peso is cryptographically accounted for."*

ACT II: Mobile Donor Application Walkthrough

Presenter Objective: Prove the mobile app's ease of use, instant cashless checkout, 5-second auto-sync, cryptographic proof, and the granular transparency of 'My Contributions'.

Scene 2.1: Authentication & Fluid Dashboard

■■ LIVE SCREEN DEMO ACTION:

Screen: Mobile App Splash Screen transitioning to Home Screen.
Action: Log in using ikathpie3@gmail.com. Scroll up and down on the Home tab to show smooth bouncing momentum physics and mouse drag scrolling. Point to **TOTAL DONATED: ■160.00** and **TRANSACTIONS: 7**.

■■ SPEAKER SPOKEN SCRIPT:

"We begin with the mobile donor experience. Parishioners authenticate with secure JWT tokens. On the Home Screen, parishioners receive a clean summary of their lifetime contributions—here showing ■160.00 across 7 transactions. Notice how our interface implements smooth bouncing scroll physics and mouse drag support, ensuring an intuitive, native mobile feel."

Scene 2.2: Automated Cashless Donation via PayMongo

■■ LIVE SCREEN DEMO ACTION:

Screen: Bottom Navbar → Tap the centered **Donate** button.
Action: Choose **'Parish General Fund'** (or **'Typhoon Calamity Aid'**). Select **■25.00**. Tap **'Proceed to PayMongo Checkout'**. Display the simulated QR Ph / GCash payment window.

■■ SPEAKER SPOKEN SCRIPT:

*"Now, let's perform a live donation. Donors don't just give blindly—they select a specific destination sector, such as Calamity Aid or Student Scholarships. By integrating **PayMongo**, we eliminate physical cash handling risks. The donor scans a standard QR Ph code using GCash, Maya, or any banking app. Once paid, our webhook automatically updates our backend in real time."*

Scene 2.3: 5-Second Real-Time Auto-Refresh & Blockchain Hash

■■ LIVE SCREEN DEMO ACTION:

Screen: Tap the **History** tab on the bottom navbar.
Action: Show the donation list. Explain that it auto-refreshes every 5 seconds silently without a loading spinner. Tap to expand the top donation card (■25.00). Scroll down inside the card and point to the **Blockchain Information** box with the 0x... hash. Tap the **Copy** button to show the toast notification.

■■ SPEAKER SPOKEN SCRIPT:

"Navigating to the History tab, observe that the screen silently refreshes every 5 seconds—any new receipt or status change appears instantly without reloading the page.

*When we expand this receipt, notice the **Blockchain Information** box containing a hexadecimal hash starting with 0x. This is not a dummy ID—it is the cryptographic transaction receipt mined into our **Hyperledger Besu** smart contract. It mathematically proves that sto. Domingo parish officially logged this ■25.00 into their immutable ledger."*

Scene 2.4: Reports & Transparency — 'My Contributions' Impact

■■ LIVE SCREEN DEMO ACTION:

Screen: Tap **Donation Transparency** (or open Transparency Screen from Home).
Action: Scroll down to show the overview card collapsing smoothly while the 3 tabs pin at the top. Focus on the **'My Contributions'** tab. Point to the progress bar: **Disbursed: ■45.00 (28%)** vs **Cash on Hand: ■115.00 (72%)**. Tap open the card for ■25.00 to show **GIVING IMPACT BREAKDOWN**: *'Community Kitchen Feeding Program'*, ■25.00, notes: *'Purchased warm meals for 15 children'*.

■■ SPEAKER SPOKEN SCRIPT:

*"This brings us to the centerpiece of our mobile app: the **Reports & Transparency** module.*

*Under the newly refined **'My Contributions'** tab, donors can clearly track their money's lifecycle. The progress gauge illustrates that out of ■160.00 given, ■45.00 has been deployed to missions, while ■115.00 is safely held in the treasury.*

*Even better, expanding the card reveals the **Giving Impact Breakdown**: the donor sees that their ■25.00 was disbursed for the 'Community Kitchen Feeding Program' on October 14 to feed 15 children. This fulfills our core thesis: complete, itemized post-donation accountability."*

ACT III: Web Administrative & Prescriptive Portal Walkthrough

Presenter Objective: Demonstrate how parish administrators manage inflows, allocate and disburse aid, utilize prescriptive analytics, and maintain financial integrity.

Scene 3.1: Admin Dashboard & Financial Solvency

■■ LIVE SCREEN DEMO ACTION:

Screen: Switch display to Chrome Browser showing React Web Portal (localhost:3000).
Action: Show the Superadmin navigation bar and point to top KPI cards: Total Donated Pool, Actual Disbursed, and Treasury Reserves.

■■ SPEAKER SPOKEN SCRIPT:

"Now, we transition to the parish administrative portal. This is where parish clergy and financial officers manage the entire operation. The Executive Dashboard aggregates parish collections, active beneficiary demographics, and historical relief spending."

Scene 3.2: Disbursing Funds & Generating Mobile Usage Logs

■■ LIVE SCREEN DEMO ACTION:

Screen: Click **Donation Management** on the sidebar.
Action: Click on an approved donation (e.g., ■30.00 General Fund). In the modal, click **Edit Status / Disburse**. Select status: **'partially used'**, enter Disbursed Amount: **20.00**, Purpose: **'Typhoon Relief Operation'**, Notes: **'Purchased 5 relief food packs for affected families'**. Click **Save**.

■■ SPEAKER SPOKEN SCRIPT:

"When the parish conducts an outreach mission, the administrator opens the verified donation record. Here, they disburse funds: setting the status to 'Partially Used', entering ■20.00 for the Typhoon Relief Operation, and writing operational notes. The system calculates the remaining balance of ■10.00 automatically. The moment this is saved, our backend updates the donor's mobile screen instantly."

Scene 3.3: Prescriptive Analytics & Optimization Matrix

■■ LIVE SCREEN DEMO ACTION:

Screen: Click **Reports** on the sidebar to reveal the Prescriptive Decision Support table.
Action: Point out the columns: **Ministry Sector**, **Urgency Multiplier** (e.g. 1.6x for Scholars, 1.4x for Feeding), **Headcount**, **Target Share**, **Actual Disbursed**, and **Variance / Gap**.

■■ SPEAKER SPOKEN SCRIPT:

*"A standout feature of ReliefLink is our **Prescriptive Decision Support Engine**. Instead of guessing how to distribute donations, our mathematical algorithm evaluates registered beneficiaries and weights them by urgency: such as a 1.6x multiplier for student education and 1.4x for indigent nutrition.*

The system generates a Prescriptive Target and highlights remaining allocation gaps in yellow. This guides parish councils to allocate resources scientifically where the need is highest."

ACT IV: Hyperledger Besu Blockchain & Tamper-Evidence

Presenter Objective: Prove the technical credibility of the blockchain layer. Explain why Hyperledger Besu is used and show how immutability guarantees audit compliance.

■■ LIVE SCREEN DEMO ACTION:

Screen: Show terminal or Besu explorer logs showing block generation or open DonationRegistry.sol in VS Code.
Action: Highlight function recordDonation(...) with donor address, amount, reference number, and timestamp.

■■ SPEAKER SPOKEN SCRIPT:

*"Underpinning our architecture is our private **Hyperledger Besu Ethereum blockchain** running smart contract DonationRegistry.sol.*

Why Besu? Unlike public Ethereum which suffers from unpredictable gas fees, Besu is an enterprise-grade consortium chain. Parishes and dioceses can operate validator nodes with zero gas costs (`min-gas-price = 0`).

Every approved donation mints an immutable block. If an insider or hacker tampers with the MongoDB database to inflate numbers or fabricate disbursements, our verification script flags the hash mismatch immediately. The blockchain acts as an impartial, mathematical witness."

ACT V: Conclusion & Anticipated Defense Questions

Presenter Objective: Deliver a crisp, memorable closing statement, then refer to the quick-defense cheat sheet for panel questions.

■■ LIVE SCREEN DEMO ACTION:

Screen: Show split-screen of Mobile App alongside Web Dashboard, or conclusion slide.
Action: Both presenters stand, thank the panel, and transition to Q&A.;

■■ SPEAKER SPOKEN SCRIPT:

"To summarize, ReliefLink proves that transparency and modern technology can coexist to elevate donor trust. By merging automated mobile cashless donations, granular post-donation disbursement tracking, and private blockchain immutability, we transform charity from a leap of faith into an audited, verifiable partnership between parishioners and the church.

Thank you very much. We are now eager to answer your questions."

Panel Q&A; Defense Strategy Sheet

Anticipated Panel Question	Recommended High-Scoring Technical Answer
1. Why did you use blockchain when a standard database like MongoDB already stores transactions?	A traditional database has root administrators who can perform silent UPDATE or DELETE queries to hide anomalies. A blockchain is append-only: once a block hash is computed and linked to prior blocks, altering any past entry invalidates the entire chain, guaranteeing tamper evidence.
2. Do parishioners need crypto wallets like MetaMask to donate?	No. Parishioners donate using familiar Philippine payment channels (GCash, Maya, QR Ph via PayMongo). The blockchain layer runs transparently in the background on private Hyperledger Besu nodes without exposing crypto complexity to users.
3. How does the system handle canceled or rejected donations?	Canceled donations never reach treasury deposit status. In our mobile 'My Contributions' view, canceled records are filtered out completely so they never skew financial outflow metrics or disbursement balances.
4. How does the mobile app stay updated when admin disburses funds?	We implemented a periodic background polling architecture (5-second silent sync) in Flutter that pulls verified usage logs without causing UI stuttering or scroll resets.